

One Wrong Move: Lower-Is-Better Scoring Report

Final summary: The scoring flow uses a clear time-score: rounded-up active solving seconds plus a 10-second penalty per mistake. The UI, share text, scoring utility, validations, README, and design documentation are present on branch `scoring-lower-is-better`. Browser verification passed, including timer pause behavior, mistake counting, completion breakdown, share text, Variant 2 replay completion, and no observed console/page errors.

Repo State Summary

- Repository: `/Users/kevinhegg/Desktop/one-wrong-move`
- Branch at start: `scoring-lower-is-better`
- Remote: `origin git@github.com:KevinHegg/one-wrong-move.git`
- Initial worktree before this verification: clean.
- Checkpoint commit: skipped because there were no uncommitted puzzle-pool/paused-flow files.
- Existing branch commit found: `959e302 Change scoring to lower-is-better time score`, already on origin before this report refresh.

Files Created Or Changed

- `public/scoring.js`: browser/Node scoring utility.
- `scripts/validate-scoring.js`: exact scoring case validation.
- `public/index.html`: loads scoring before game and includes result breakdown fields.
- `public/game.js`: uses `calculateScore`, new completion display, and new share text.
- `public/style.css`: compact score breakdown styling.
- `README.md` and `docs/game-design.md`: scoring formula, examples, active-time timer note, mistake penalty rationale.
- `docs/codex-reports/screenshots/006-*.png`: fresh browser captures.
- `docs/codex-reports/006-lower-is-better-scoring-report.pdf`: this report.

Old Scoring Problem

The old formula produced abstract high-looking values such as `934`. That read like points and made it unclear whether higher or lower was better.

```
score = max(0, 1000 - floor(seconds * 10) - mistakes * 50)
```

New Scoring Formula

```
baseSeconds = Math.ceil(totalActiveMs / 1000)
mistakePenaltySeconds = mistakes * 10
scoreSeconds = baseSeconds + mistakePenaltySeconds
```

The timer measures active solving time only. Mistakes add time so inaccurate guesses matter without introducing abstract point totals. The 10-second penalty is currently blunt, compact, and easy to understand.

Worked Examples

- `0ms`, 0 mistakes: base 0s, penalty 0s, score 0s.
- `1000ms`, 0 mistakes: base 1s, penalty 0s, score 1s.
- `1001ms`, 0 mistakes: base 2s, penalty 0s, score 2s.
- `18200ms`, 2 mistakes: base 19s, penalty 20s, score 39s.
- `59999ms`, 3 mistakes: base 60s, penalty 30s, score 90s.

Validation Results

- `node --check public/puzzles.js`: passed.
 - `node --check public/game.js`: passed.
 - `node --check public/scoring.js`: passed.
 - `node scripts/validate-scoring.js`: Validated lower-is-better scoring cases.
 - `node scripts/validate-puzzles.js`: Validated 1200 daily rounds across 100 dates and 7 puzzle types.
 - `git diff --check`: passed.
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Browser Verification

- Intro explains: Score = rounded-up solving time + 10s per mistake. Lower is better.
 - Timer stayed at 0.0s in intro/briefing and started after **Start Round**.
 - Wrong tap increased the visible mistake count and left the timer running.
 - Correct tap paused the timer and disabled the solved board.
 - Completion showed Solved 3 of 3, active time, base time, mistake count, penalty calculation, final score in seconds, and Lower is better.
 - Share text used the new score format and included round names.
 - Play another mix worked and Variant 2 completion/share text included Variant 2.
 - Browser reload watch observed no console events and no page errors.
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Local Run Instructions

```
python3 -m http.server 8080 -d public
# then open http://127.0.0.1:8080/
```

In this run, port 8080 was already occupied by an existing Python SimpleHTTP server serving the app, so that listener was reused.

Git Commands Run

```
pwd
git status --short
git branch --show-current
git remote -v
find . -maxdepth 3 -name .git -type d
ls -la
node --check public/puzzles.js
node --check public/game.js
node scripts/validate-puzzles.js
git diff --check
git switch -c scoring-lower-is-better || git switch scoring-lower-is-better
node --check public/scoring.js
node --check public/game.js
node scripts/validate-scoring.js
node scripts/validate-puzzles.js
git diff --check
python3 -m http.server 8080 -d public
```

Git Commands For User

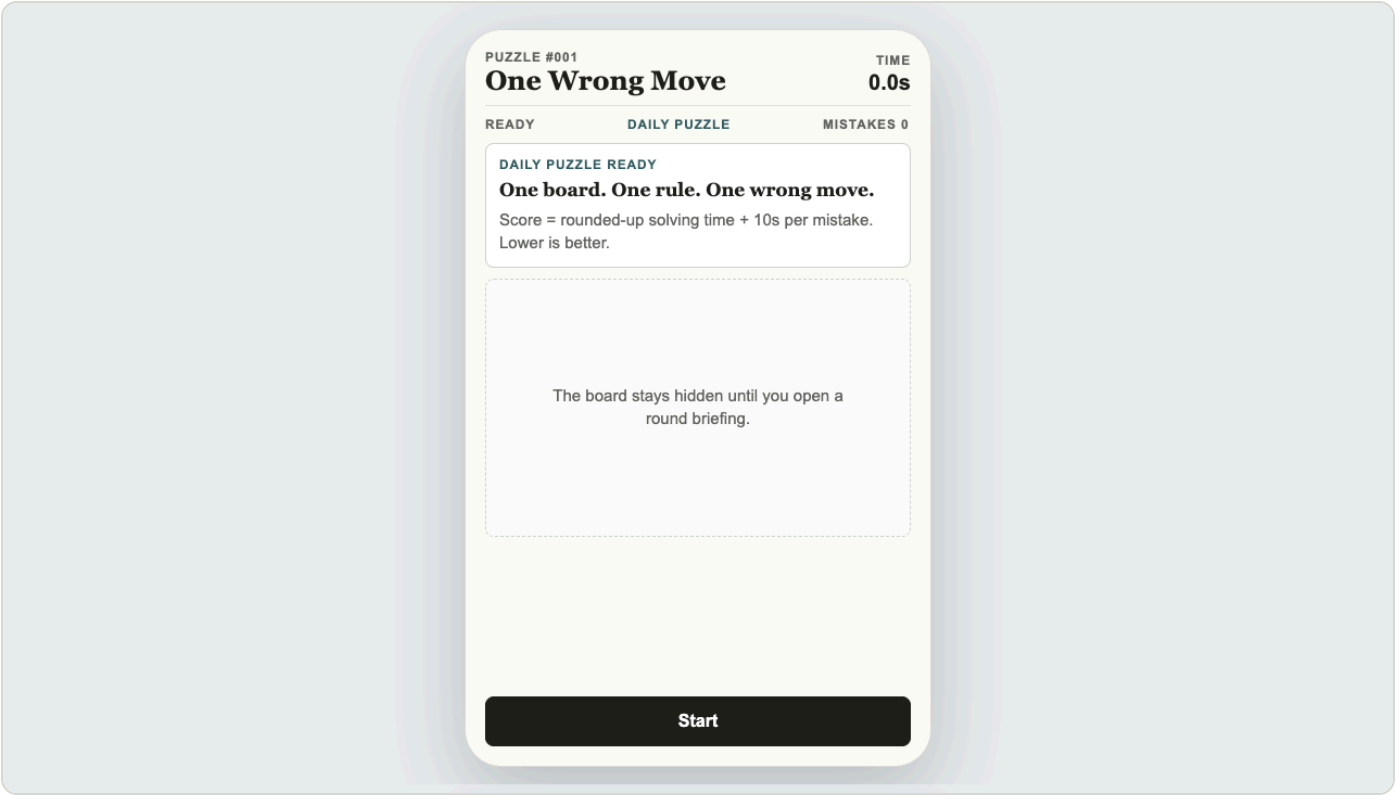
No manual Git command is expected if the final push succeeds. If pushing cannot be completed, run:

```
git push origin scoring-lower-is-better
```

Screenshots

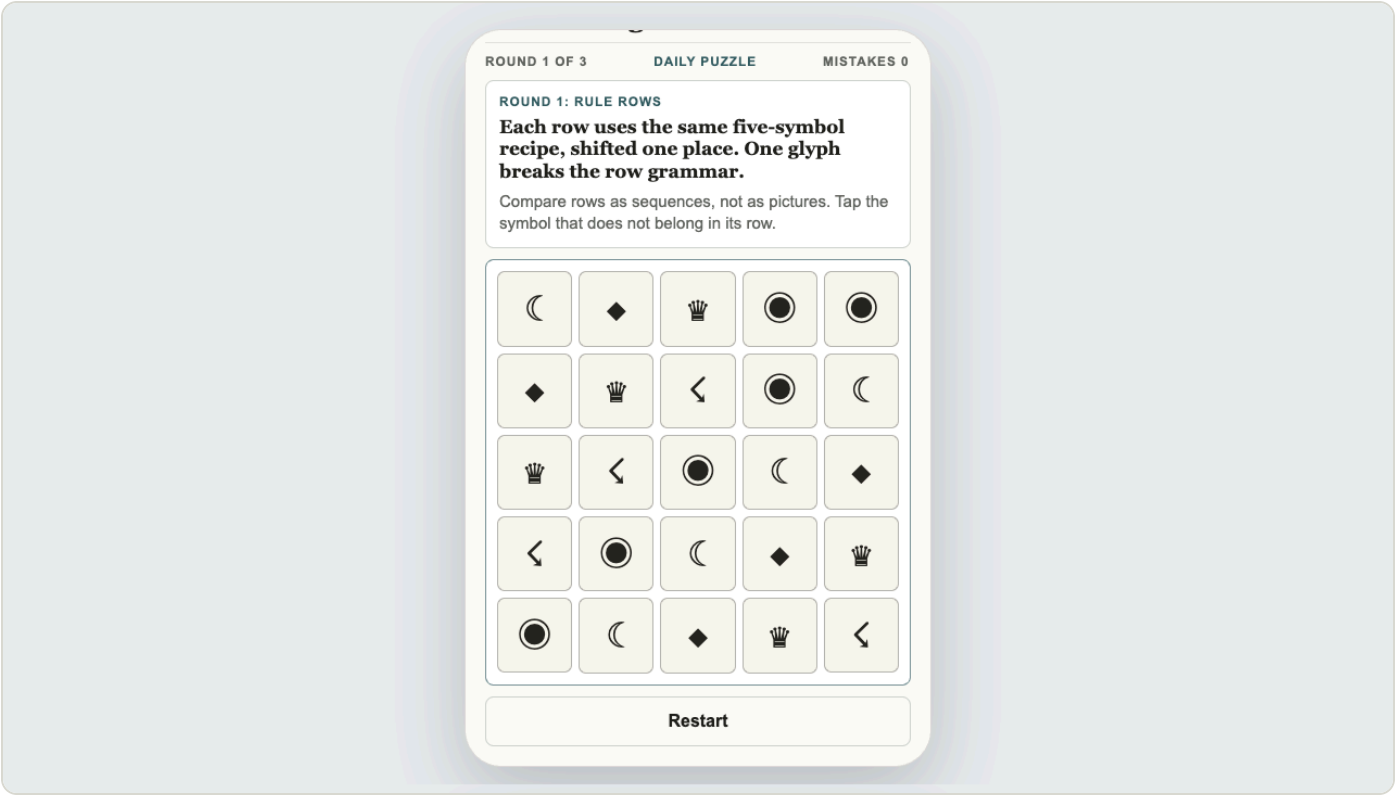
Intro screen showing scoring explanation

006-intro-scoring-explanation.png



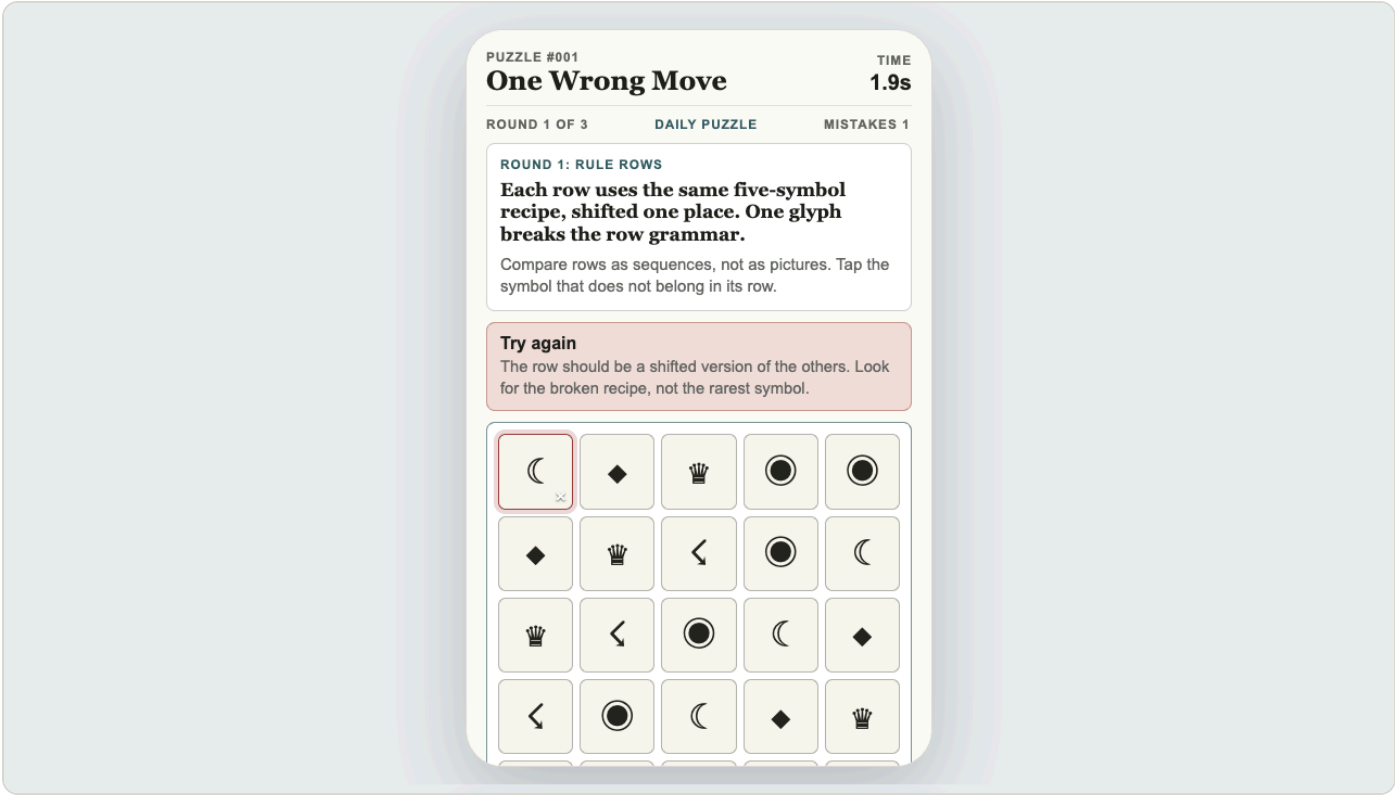
Active round screen showing timer still works

006-active-round-timer.png



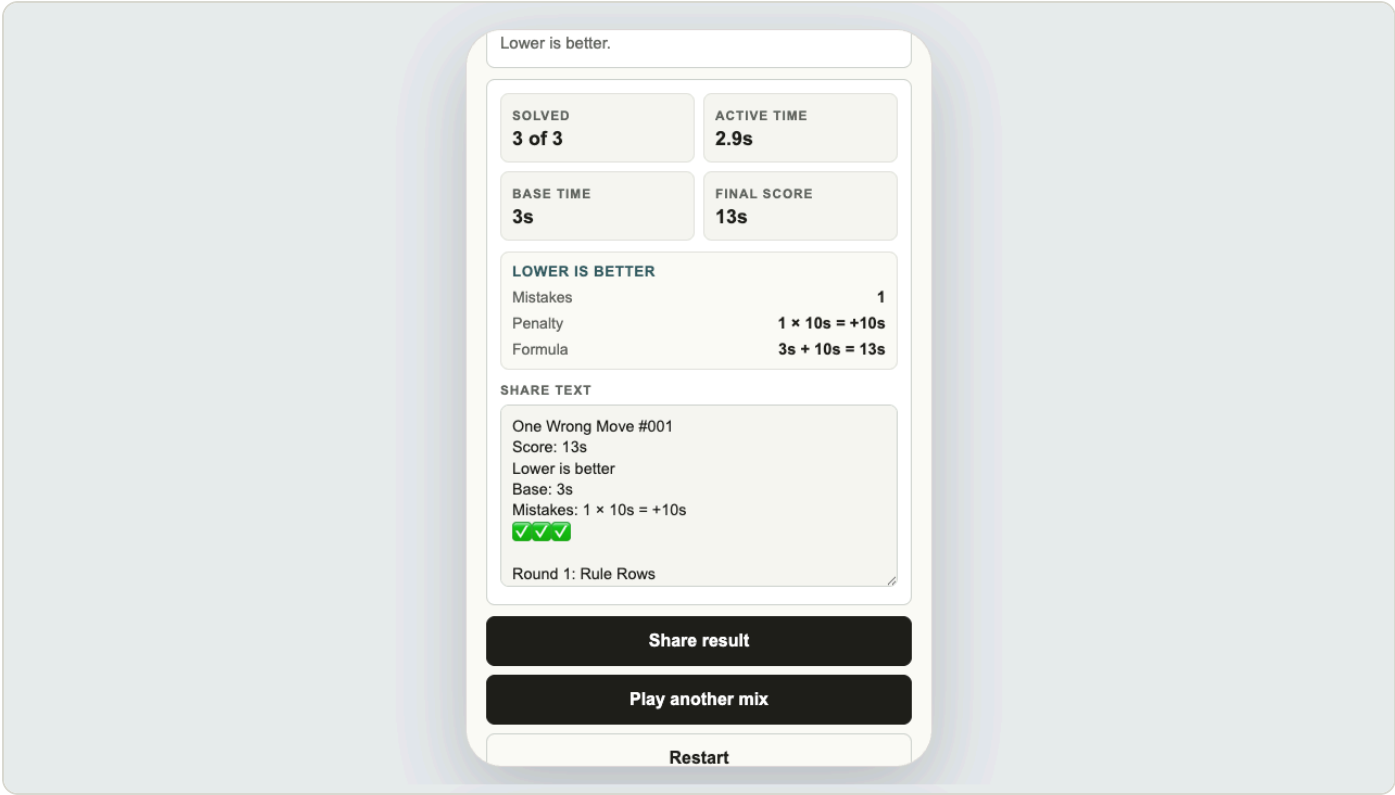
Wrong-tap state showing mistake count

006-wrong-tap-mistake-count.png



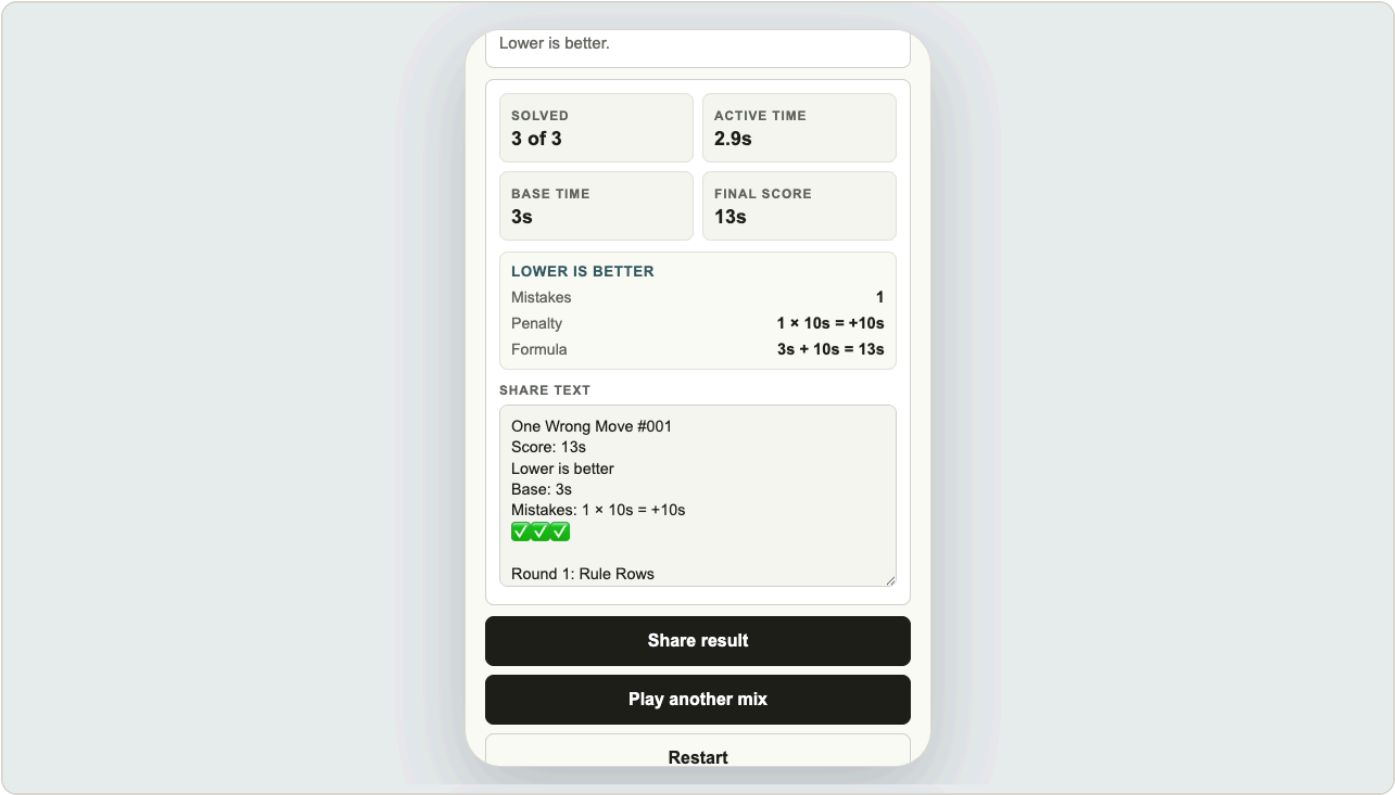
Completion screen showing score breakdown

006-completion-score-breakdown.png



Share text with lower-is-better score

006-share-text-new-score.png



Variant/replay completion with Variant 2 share text

006-variant-replay-completion.png

